Universiti Tun Hussein Onn Malaysia

UNIVERSITI TUN HUSSEIN ONN MALAYSIA

FINAL EXAMINATION
SEMESTER I
SESSION 2022/2023

COURSE NAME	:	DISCRETE STRUCTURE
COURSE CODE	:	BIC 10103
PROGRAMME CODE	:	BIS/ BIP/ BIW/ BIM
EXAMINATION DATE	:	FEBRUARY 2023
DURATION	:	3 HOURS
INSTRUCTION	:	1. ANSWER ALL QUESTIONS. 2. THIS FINAL EXAMINATION IS CONDUCTED VIA **CLOSED BOOK.** 3. STUDENTS ARE **PROHIBITED** TO CONSULT THEIR OWN MATERIAL OR ANY EXTERNAL RESOURCES DURING THE EXAMINATION CONDUCTED VIA CLOSED BOOK.

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THIS QUESTION PAPER CONSISTS OF **FOUR (4)** PAGES

Q1 (a) Suppose that a function, f is defined recursively by

$$f(0) = 2$$
$$f(n+1) = 3f(n) - 1$$

Find the first five terms of the function.

(5 marks)

(b) Use mathematical induction to prove that $5^n + 3$ is divisible by 4 for all positive integer n.

(10 marks)

Q2 (a) Show that the relation R in the set $A = \{1,2,3,4,5\}$ given by $R = \{(a,b): |a - b| is\ even\}$ is an equivalence relation. Show that all the elements of $\{1,3,5\}$ are related to each other and all elements of $\{2,4\}$ are related to each other. But no elements of $\{1,3,5\}$ is related to any elements of $\{2,4\}$.

(10 marks)

(b) Let $f: \{1,3,4\} \rightarrow \{1,2,5\}$ and $g: \{1,2,5\} \rightarrow \{1,3\}$ be given by $f = \{(1,2), (3,5), (4,1)\}$ and $g = \{(1,3), (2,3), (5,1)\}$.

Find out the value of $g \circ f$.

(5 marks)

Q3 (a) A person deposits RM1,000 in an account that yields 9% interest compounded annually. Set up a recurrence relation for the amount in the account at the end of n years and how much will be in the account after 10 years.

(5 marks)

(b) Formulate a solution to the following recurrence relations.

$a_n = 5a_{n-1} - 6a_{n-2}$ where $a_0 = 1, a_1 = 0$.

(8 marks)

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Q4 (a) Consider the following pseudocode segment.

```
1    y = x + x + x + x + x
2    z = y + y + y
```

(i) Identify the number of operations in the pseudocode segment.

(2 marks)

(ii) If $x = 3$ before the segment is executed, identify the value of z after execution.

(2 marks)

(b) Consider the following pseudocode segments. Show the number of words that each algorithm print.

```
for i = 1 to 9
    for j = 1 to 6
        for k = 1 to 3
            print "she wins"
```

(3 marks)

(c) Consider the following algorithm.

```
Procedure linear search (x: integer, a1, a2,…,an: distinct
integer)
i = 1
while (i ≤ n and x ≠ ai)
      i = i + 1
if i ≤ n then location = i
else location = 0
return location {location is the subscript of the term
equals x, or is 0 if x is not found}
```

(i) Suppose that a_i = {21, 9, 13, 27} and the algorithm requires to locate an element of 27. Calculate the number of comparisons made by this algorithm.

(6 marks)

(ii) Determine the time complexity of the algorithm.

(2 marks)

(iii) Find the big-O estimation for the worst-case analysis of the algorithm.

(2 marks)

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Q5 (a) Answer **Q5(a)(i) – Q5(a)(ii)** based on **Figure 5(a)**.

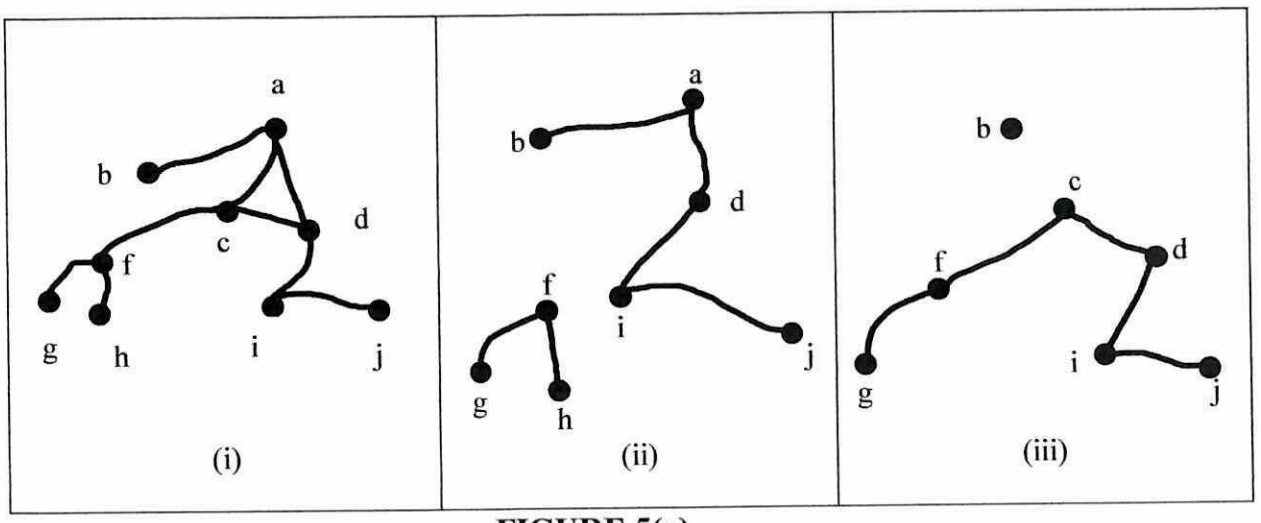

FIGURE 5(a)

(i) How many connected subgraphs which have four vertices and include a path?
(1 mark)

(ii) How many spanning subgraphs are there for the graph (i).
(2 marks)

(b) Draw the subgraph of G induced by the set of vertices U = $\{b, c, d, f, i, j\}$
(5 marks)

(c) If G is the graph in **Figure 5(b)**, the edges $\{a,b\}\{b,c\}\{c,f\}\{f,e\}$ $\{e,d\}\{d,g\}\{g,h\}\{h,i\}$ is Hamilton path for G. Explain whether G have a Hamilton cycle or not.

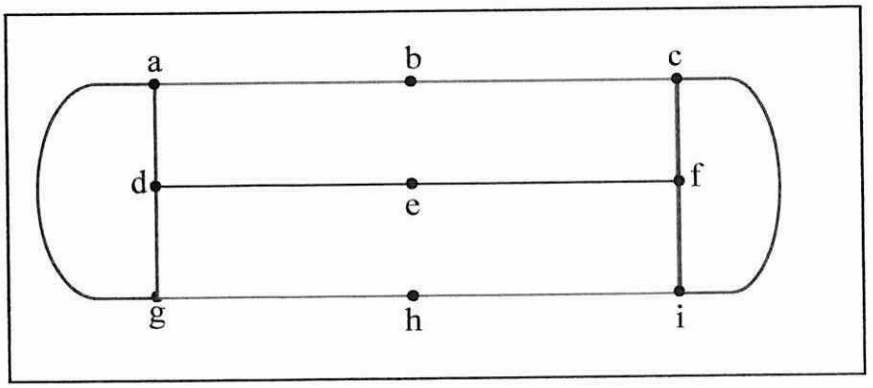

FIGURE 5(b)

(12 marks)

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- END OF QUESTIONS –